PARAM SHARMA

AI/ML Engineer

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PROFESSIONAL SUMMARY

Computer Science graduate and **AI/ML Engineer** specializing in **Computer Vision**, **LLM fine-tuning**, and **end-to-end MLOps architectures**. Authored multiple peer-reviewed publications and open-source datasets, bridging the gap between state-of-the-art research (Swin Transformers, custom optimization) and production deployment. Proficient in PyTorch, TensorFlow, **Python**, **C++**, and cloud native **MLOps** tools, dedicated to shipping scalable high-accuracy AI systems.

WORK EXPERIENCE

AI/ML Engineer | Full-time — May 2025 - Present
MaxBrain Technologies — *On-site • Jaipur, India*

Plant Disease Detection System (FasalSaathi)

- **Curated and published** two open-source agricultural image datasets for **Ridge Gourd** pathology (IEEE DataPort, 9,000+ downloads) and **Brinjal pathology** (Mendeley Data).
- Benchmarked deep learning architectures (VGG19, ResNet, InceptionNet) and engineered a hybrid pipeline using a **pre-trained DenseNet121** as a feature extractor paired with an **XGBoost classifier**, boosting accuracy **from 78% (baseline CNN) to 92%.**
- Deployed the optimized inference model into **"FasalSaathi,"** a live Android application on the Google Play Store, and secured an **Indian Design Patent** for an IoT-based automated agricultural data collection device.

Smart EdTech Platform (MCQGyaan AI)

- **Engineered** an enterprise EdTech platform featuring an **AI-driven evaluation engine** capable of semantically grading handwritten and typed answers using **OCR and NLP**.
- **Curated** custom educational datasets to fine-tune **Large Language Models (LLMs)** for context-aware assessment, significantly reducing educator grading time.
- **Developed** full-stack, **role-based portals** featuring automated question paper generation and real-time student performance analytics.

Technologies / Skills Used: *Python, PyTorch, CNN, LLM Fine-Tuning, NLP, OCR, REST API, React, MongoDB*

PROJECTS

Cloud-Native MLOps & Observability Pipeline | GitHub

- **Architected** an end-to-end NLP MLOps pipeline, orchestrating multi-stage DAG workflows with **DVC and AWS S3** for robust data and artifact versioning.
- **Integrated MLflow and DagsHub** for experiment tracking, alongside automated CI/CD quality gates via **GitHub Actions** for programmatic model promotion.
- **Containerized** and deployed the inference API to an **AWS EKS (Kubernetes)** cluster, implementing custom **Prometheus** instrumentation and **Grafana** dashboards for live observability.

Technologies / Tools Used: *AWS EKS, MLflow, DVC, Prometheus, Grafana, Docker, GitHub Actions, NLP*

Brain Tumour Classification (Swin-GWO-Net++) | GitHub

- Developed the **Swin-GWO-Net++ hybrid transformer** model as a final-year Bachelor's thesis, achieving **99.54% classification accuracy** across 102,600 BraTS 2021 brain MRI scans.
- Engineered a **Dual Attention Fusion Module (DAFM)**, resulting in **a 2.74% accuracy** increase over standard backpropagation in ablation studies.
- Implemented **Grad-CAM++ feature mapping to provide visual explainability** and clinical interpretability for deep learning model predictions.

SKILLS

Programming Languages:	C++, Java, Python, JavaScript, TypeScript, SQL
Frameworks & Libraries:	Django, Flask, FastAPI, Next.js, React, Tailwind CSS
AI, LLMs and Deep Learning:	PyTorch, TensorFlow, Scikit-learn, OpenCV, RAG, Agentic AI, LLM Engineering & Fine-Tuning, Hugging Face,vLLM, Unsloth, LangGraph
Tools, MLOps & Automation:	AWS, Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, Prometheus, Grafana, n8n, Model Deployment Streamlit, Git, GitHub, MongoDB, PostgreSQL

EDUCATION

Birla Institute of Technology Mesra | Ranchi — Nov 2022 - Jun 2026
B.Tech CSE — *CGPA: 8.3/10*
Major in **Computer Science** and **Engineering**

University of Tokyo | Japan — Apr 2026 - Jul 2026
Matsuo Lab GCI Program
Selected for the Global Creator Intensive (GCI) — a 3-month AI/ML and Data Science program at Matsuo Laboratory, University of Tokyo. Covered deep learning theory, large-scale array computation, time-series analysis, unsupervised learning, and participated in global data competitions.

AWARDS & ACHIEVEMENTS

RESEARCH & PUBLICATIONS

- **Authored 2 Peer-Reviewed AI/ML Research Papers:** Published main author research at **WCAIAA 2026** (TinyML MobileNet1D quantized to <1.5 MB with 92.77% accuracy) and co-authored plant disease diagnosis research in **Springer AIIS (MIND 2025)**.
- **Published Open-Source Datasets:** Created and released the *Ridge Gourd Pathology Dataset* on **IEEE DataPort** (9,000+ downloads, 335+ ratings, featured in IEEE Aug 2025 Newsletter) and the *Brinjal Pathology Dataset* on **Mendeley Data**.
- **Intellectual Property (IP):** Granted a **Government of India Design Patent** for an IoT-based automated agricultural data collection device.